Ricardo Carreira da Silva

Mechanical Engineer

Oliveira de Azeméis, Portugal ricardo.carreira.silva@gmail.com
linkedin.com/in/ricardocarreiradasilva

PROFILE

Mechanical engineer with an integrated master's degree from the University of Coimbra (Production and Design branch). Experience in CAD design and prototyping, structural sizing, bills of materials and equipment budgeting. Keen interest in digital fabrication: 3D printing and laser engraving and cutting.

PROFESSIONAL EXPERIENCE

Jan 2024 – present

Mechanical Engineer · Project Manager

BTL – Indústrias Metalúrgicas, S.A. · Ossela, Oliveira de Azeméis, Portugal

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Feb 2023 – present

Mechanical Engineer

BTL – Indústrias Metalúrgicas, S.A. · Ossela, Oliveira de Azeméis, Portugal

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Nov 2021 – Feb 2023

Machine Setter

Massilly Portugal · Portugal

- Assembled and set up press-shop equipment and machines
- Documented machine set-up procedures
- Took part in continuous improvement projects

Apr 2021 – Jun 2021

Census enumerator, 2021 Census

Instituto Nacional de Estatística (INE) · Portugal

- Planned fieldwork with the parish coordinator and the municipal delegate
- Checked the completeness of buildings, dwellings and subsections
- Distributed code letters and matched them to dwellings
- Helped residents complete the census questionnaires
- Collected data in the assigned area

Jan 2018 – Nov 2018

Engineering Intern (Industrial and Production)

MCM – Moreira, Cruz e Magalhães, Lda. · Vilar do Pinheiro, Vila do Conde, Portugal

- Built prototypes in AutoCAD and CATIA V5
- Structural sizing and problem analysis
- Prepared bills of materials (BOM)
- Selected materials for prototype construction
- Revised prototypes and developed better-value alternatives within the project budget
- Planned and budgeted equipment for the "Fluorinated Gases" certification
- Drew schematic diagrams for swimming-pool plant rooms

EDUCATION

Oct 2008 – Jun 2016

Integrated Master's in Mechanical Engineering

Faculdade de Ciências e Tecnologia da Universidade de Coimbra (FCTUC) · Production and Design branch

- Dissertation: "Optimisation of Friction Stir Processing (FSP) of the AA5083 Alloy", supervised by José Domingos Moreira da Costa and Altino Loureiro
- Main courses: Industrial Robotics, Mechanical Technology, Industrial Automation, Industrial Control, Computational Structural Analysis, Operations Research, Technical Drawing, Mechanical Systems Design
- Took part in several extracurricular activities, especially student associations

COURSES AND CERTIFICATES

Jul 2022

First Aid for Children and Young People (training unit 9646)

Adrimag · 25 h

Apr 2022

Manual Handling of Loads

Cambra Vida · 4 h

Oct 2019 – Dec 2019

Accounting Technician Training Course

IEFP · 175 h

Dec 2017

CATIA V5 Fundamentals

ASIDEK · 40 h

Sep 2015

Plant Protection Products Applicator Course

25 h

SKILLS

Software

CATIA V5 · Autodesk Inventor · SolidWorks · AutoCAD · Ultimaker Cura · Simplify3D · Microsoft Office · Visual Basic · ADINA · SIMUL8

Technical skills

3D modelling and prototyping · Structural sizing · Cost estimating · Bills of materials (BOM) · Press equipment set-up · Continuous improvement

Personal skills

Teamwork · Leadership · Flexibility · Entrepreneurship · Initiative · Interpersonal skills · Event organisation · Willing to travel internationally

Languages

Portuguese Native **English** B2 understanding · B1 speaking and writing

Spanish A2 understanding · A1 speaking and writing

Driving licence

categories B and B1; tractors (category III)

OTHER ACTIVITIES

- Took part in the "BEST Coimbra Engineering Weeks 2013"
- Attended the conference series organised by NEEMAAC
- Volunteer at the Mechanical Engineering Students' Association of the Coimbra Academic Association (NEEMAAC), Culture department (2011)
- Seasonal farm work (fruit picking), "Colina dos Monges", summers 2008–2015

Interests: 3D printing, Laser engraving and cutting

Updated on 9 Nov 2026